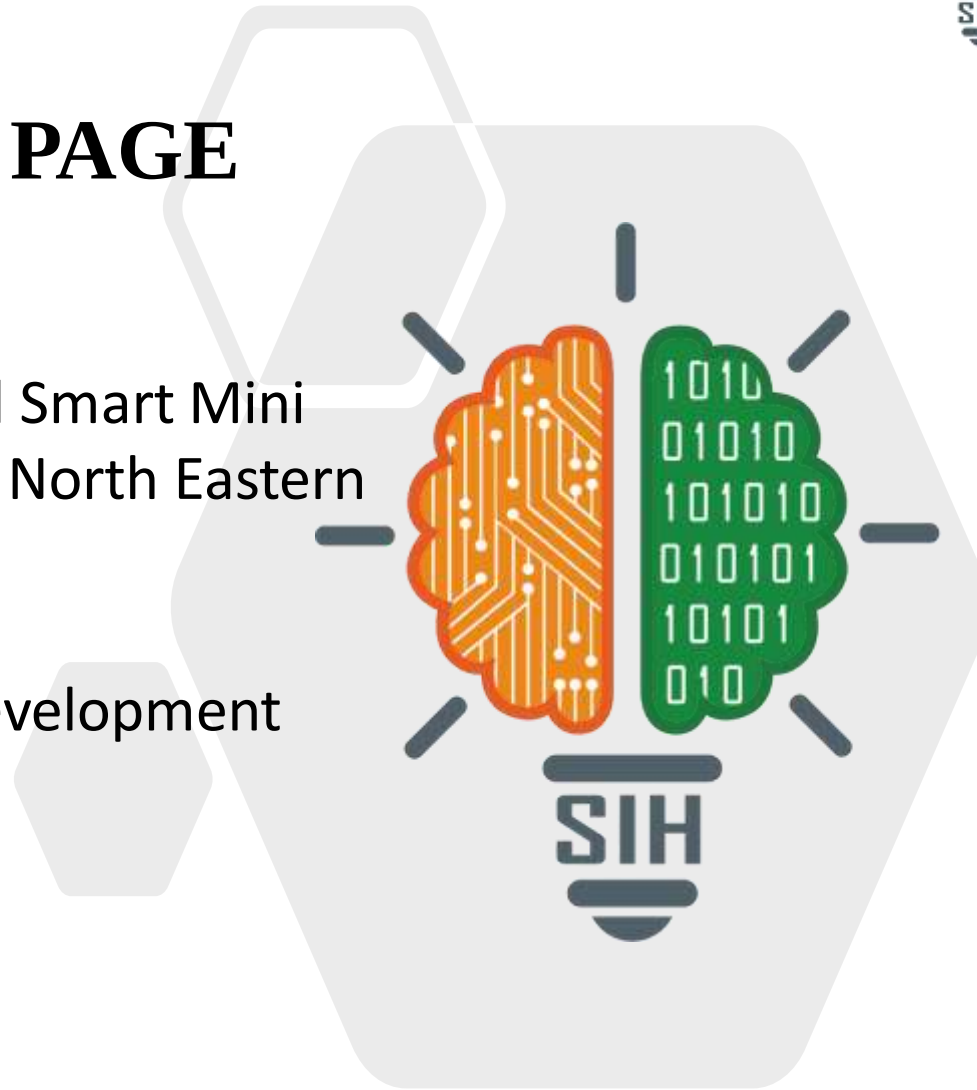


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** –SIH26005
- **Problem Statement Title** – Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER)
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Hardware
- **Team ID** –168867
- **Team Name** – LUMORA6



PROBLEM AT HAND

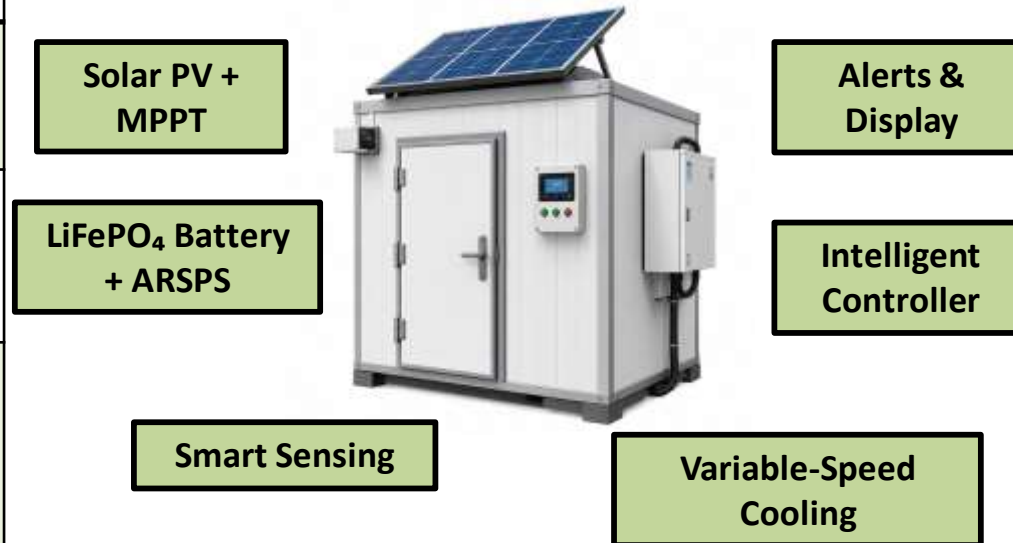
High Post-Harvest Loss: 30–40% of vegetables & fruits spoil due to lack of affordable, reliable cold storage near farms.
Unreliable Power Supply: Frequent power cuts in rural & remote NER locations make electric cold storage impractical.
High Operating Cost: Diesel-based or grid-based cold storages have high running cost and are not feasible for small farmers.
No Real-time Monitoring: Farmers lack visibility on temperature, humidity and storage conditions; silent spoilage occurs.
Battery & Energy Risks: Improper battery use leads to low backup time, high maintenance and safety risks.

IDEAL FOR

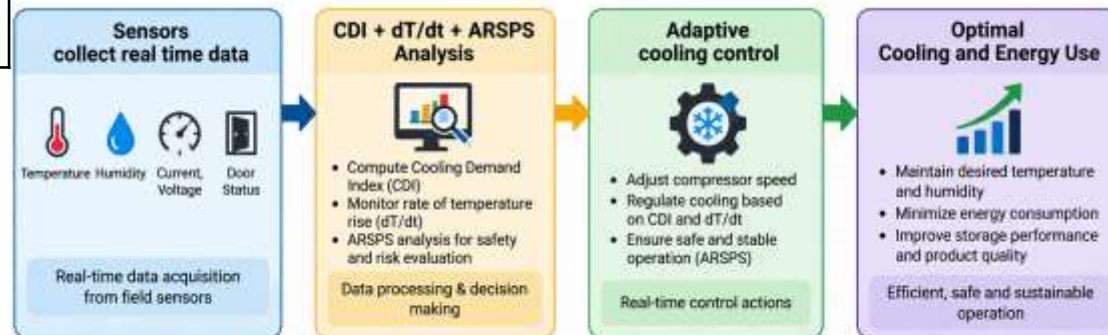
- Farm & Village Collection Centres
- Hilly & Remote NER Regions
- Small & Marginal Farmers
- Highly Perishable Produce

OUR SOLUTION

Solar-Powered Adaptive Smart Mini Cold Storage
 Decentralized • Energy-Efficient • Farmer-Friendly



HOW IT WORKS?



WHERE WE STAND OUT

Produce-Aware Shelf-Life Prediction Uses produce type, temperature & humidity history to estimate storage life for better planning and reduced waste.

Predictive Thermal Control (dT/dt) dT/dt-based early detection enables proactive cooling before unsafe conditions, reducing losses.

Adaptive Cooling (CDI) Uses Cooling Demand Index (CDI) with dT/dt, temperature, humidity, ambient & door status for adaptive cooling.

ARSPS Protection Adaptive battery protection monitors voltage, current, temperature & SOC for safety and longer battery life.

Temperature & Humidity Control Maintains produce-specific temperature & humidity to improve quality and extend shelf life.

Smart Alerts Instant alerts for temperature deviation, low energy, door open & battery faults on local display/app.

Compact & Modular Design Easy to install, scale & maintain in remote locations with minimal infrastructure.

1. TECHNOLOGIES TO BE USED



Solar System

- Solar PV
- MPPT Charger



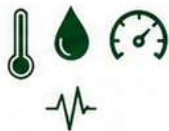
Battery & BMS

- LiFePO₄ Battery
- BMS



Cooling System

- Variable Speed DC Compressor



Sensors

- Temperature, Humidity
- Current, Voltage
- Door Status



Display & Storage

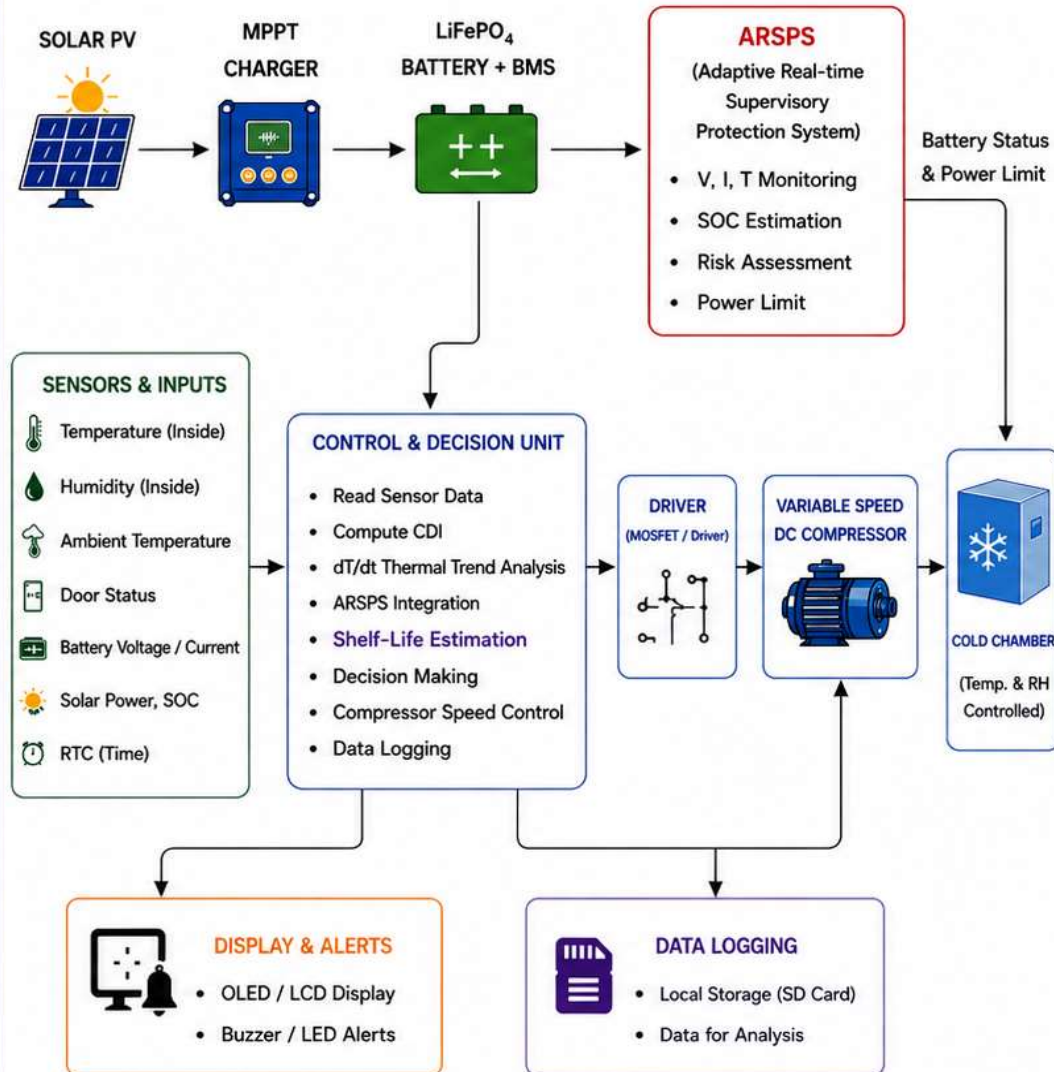
- OLED / LCD Display
- SD Card



Protection System (ARSPS)

- Voltage, Current, Temperature, SOC monitoring & Protection

2. SYSTEM ARCHITECTURE



3. METHODOLOGY & PROCESS FLOW



1 Data Acquisition

Collect all sensor data in real time.



2 Compute Indices & Predict Shelf Life

Calculate CDI (Cooling Demand Index) and estimate remaining shelf life using produce type and environmental produce type and environmental history.



3 ARSPS Analysis

Evaluate battery risk and determine safe power limit.



4 Decision Making

Controller decides optimal compressor speed based on CDI, ARSPS and shelf-life prediction.



5 Actuation

Adjust compressor speed using PWM / Driver.



6 Monitoring & Logging

Store data locally; optional cloud/mobile monitoring when connectivity is available.



7 Data-Based Optimization

Use historical operating data to tune thresholds and improve future performance.

1. ANALYSIS OF THE FEASIBILITY OF THE IDEA
Technically Feasible: Proven technologies: solar PV, LiFePO ₄ battery, variable-speed compressor and PLC control.
Economically Feasible: Estimated cost (2 MT): ₹4.5 – 5.5 lakh, comparable to the market range.
Low Manpower: Only one operator required due to automation and smart control.
High Efficiency: Estimated efficiency of 75 – 85% (20 – 30% better energy utilization).
Backup Autonomy: 36 – 48 hours of autonomy with the optimized system..
Long Life and Reliability: LiFePO ₄ battery with ARSPS protection gives 12 – 15 years of life

2. POTENTIAL CHALLENGES AND RISKS
Low Solar Availability: Reduced energy during monsoon or cloudy days.
Battery Degradation: High temperature or improper charging reduces battery life.
Sensor Errors: Inaccurate data can affect cooling decisions and efficiency.
Maintenance and Skill Gap: Lack of technical knowledge may lead to improper maintenance.
Component Failure: Compressor, PLC or sensor failure may cause downtime.
Initial Cost Sensitivity: Slightly higher initial cost may affect adoption.

3. STRATEGIES FOR OVERCOMING THESE CHALLENGES
Smart Energy Management: Use CDI, dT/dt and EAI logic to optimize cooling based on demand and available energy. Store excess energy in battery/PCM.
Battery Protection (ARSPS): Continuous monitoring of V, I, T and SOC to prevent over-stress and extend battery life.
Quality Sensors: Use reliable sensors with periodic calibration for accurate data.
Preventive Maintenance: Regular inspection, cleaning and scheduled maintenance to reduce unexpected failures.
Cost Optimization: Local sourcing, standard design and bulk procurement to reduce cost.

COST COMPARISON (2 MT MODEL)		
Solar-Powered Smart Cold Storage System		
COST COMPONENT	EXISTING SYSTEM (2 MT)	OUR PROPOSED MODEL (2 MT)
 Cold Room (PUF + Door)	₹1.20 – 1.50 L	₹1.20 – 1.50 L
 Refrigeration (Variable-speed)	₹0.65 – 0.90 L	₹0.65 – 0.90 L
 Solar PV + MPPT	₹0.55 – 0.70 L	₹0.55 – 0.70 L
 Thermal / Energy Storage (PCM TES)	₹0.50 – 0.80 L	-
 Battery (Auxiliary / Main)	₹0.10 – 0.20 L (Aux Battery)	₹0.80 – 0.95 L (LiFePO ₄ Battery)
 Control System (PLC + I/O)	₹0.25 – 0.40 L	₹0.30 – 0.45 L
 Sensors & Monitoring	₹0.07 – 0.10 L	₹0.08 – 0.12 L
 Electrical Panel & Protection	₹0.15 – 0.25 L	₹0.18 – 0.25 L
 Installation & Commissioning	₹0.30 – 0.50 L	₹0.30 – 0.50 L
 TOTAL ESTIMATED COST	₹4.0 – 6.0 LAKH*	₹4.5 – 5.5 LAKH*

* Cost may vary based on specifications, brand, and site conditions.

1.POTENTIAL IMPACT ON THE TARGET AUDIENCE	2.BENEFITS OF THE SOLUTION (SOCIAL,ECONOMIC,ENVIRONMENTAL,ETC)
FARMERS Reliable, affordable and smart cold storage ensures reduced post – harvest losses and better income.	SOCIAL BENEFITS Reduces food wastage and improves food availability. Strengthens rural livelihoods and farmer welfare. Encourages local employment for operation & maintenance.
LOCAL COMMUNITIES Better access to fresh produce, improved food security and reduced seasonal price fluctuation.	ECONOMIC BENEFITS Reduces post-harvest losses by 20-30%. Lowers operating cost through smart energy management. Increases farmer income with better produce quality and longer storage.
SMALL BUSINESSES / TRADERS Consistent quality storage enables longer storage and wider market reach.	ENVIRONMENTAL BENEFITS Uses solar energy – clean and renewable. Lower carbon footprint compared to diesel – based systems. Energy efficiency reduces overall resource consumption.
GOVERNMENT & DEVELOPMENT AGENCIES Supports rural development goals, agri-value addition and self-reliance in cold chain infrastructure.	TECHINCAL & OPERATIONAL BENEFITS Smart adaptive cooling ensures optimal temperature control. Battery safety (ARSPS) extends battery life (12-15 years). 36-48 hours backup without sun ensures reliable operation.
ENVIRONMENT & SOCIETY Promotes clean energy use and sustainable technology adoption in rural areas.	LONG –TERM IMPACT Builds resilient and self-sustained cold chain in rural areas. Scalable model for 1 MT – 10 MT and future IoT integration. Promotes sustainable agriculture and economic growth.

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- [3] W. Amjad, A. Munir, F. Akram, A. Parmar, M. Precoppe, F. Asghar, and F. Mahmood, “Decentralized solar-powered cooling systems for fresh fruit and vegetables to reduce post-harvest losses in developing regions: A review,” *Clean Energy*, vol. 7, no. 3, pp. 635–653, 2023, doi: 10.1093/ce/zkad015.
- [4] H. J. El-Khozondar, M. Elnaggar, A. Al-Mabouh, L. Abu Zanouna, M. I. Al-Atawne, and O. Safi, “Solar-powered cold storage system,” in *Proc. 2021 International Conference on Promising Electronic Technologies (ICPET)*, 2021, pp. 28–33, doi: 10.1109/ICPET53277.2021.00012.